# Indraprastha Power Generation Co. Ltd. vs Delhi Electricity Regulatory ... on 10 January, 2008

JUDGMENT

Appeal No. 81 of 2007

1. The appellant is a Government owned company having three power generating stations namely :

1. Indraprastha Thermal Power Station (IPTPS for short),

2. Rajghat Power House (RPH for short) and

3. Indraprastha Gas Turbine Power Station (IPGTPS for short).

2. The respondent No. 2, Delhi Transco Ltd. is a transmission licensee. The appellant and the respondent No. 2 entered into a Power Purchase Agreement (PPA) on 30.03.2007. On 29.03.2007, the draft PPA had been approved by Delhi Electricity Regulatory Commission (DERC or Commission for short). Vide the impugned order dated 22.09.2006, the DERC decided the ARR and tariff applicable to the appellant for generation and sale of electricity from the aforesaid three power plants. The appellant, thereafter, filed a review petition, being No. 54/2006. The review petition having been dismissed vide the other impugned order dated 30.03.2007, the appellant has preferred the present appeal. The present appeal challenges the order dated 22.09.06 in respect of the following aspects:

1. Non consideration of Operation & Maintenance (O&M) charges on actual basis,

2. Lower heat rate norms specified for the power projects,

3. Non consideration of final outflow to the appellant on account of rebate on timely payment allowed by the appellant to the respondent No. 2,

4. Fixation of 60% combined plant load factor of the RPH project for financial year 2005-06 and 2006-07 for recovery of fixed cost and

5. Lower variable cost for IPGTPS.

The challenge:

3. The contention of the appellant in respect of each factor is delineated below:

(a) O&M Expenses : The appellant had submitted an estimate of Rs. 153.51 Crores as O&M expenses for the financial year 2006-07. The Commission has allowed an

expenditure of Rs. 99.44 Crores as recoverable from revenue on normative basis as O&M expenses. The appellant contends that it had acquired the power plants from the erstwhile Delhi Vidyut Board (DVB), of which some are 38 years old and some 15-20 years old, that appellant is incurring substantial expenditure on repair and maintenance activities and has recently incurred more than Rs. 15 Crores in RPH to rectify the problem of an axial shift and high vibration which has resulted in those units having achieved 80% Plant Load Factor (PLF), that the appellant is incurring a high wage bill due to the number of employees being large, that it has given VRS to 383 employees in order to optimize the man power and that despite these efforts the actual O&M expenses for the financial year 2005-06 was Rs. 113.75 Crores. The appellant claims that the O&M expenses claimed were within the parameters of the Central Electricity Authority's (CEA) recommendations. The Commission assessed O&M expenditure for 2005-06 at Rs. 95.61 Crores & for 2006-07 at Rs. 99.44 Crores arrived at by applying an escalation factor of 4% over previous year's expenditure. According to the appellant the Commission should consider actual expenses incurred for 2005-06 and apply the escalation factor on such actual figure for arriving at the estimated O&M expenses for 2006-07

(b) Heat Rate: The heat rate submitted by the appellant for approval was as under:

STATION 2005 06 (Actual) 2006 07 (As per Petition) IPTPS 3709 Kcal/kwh 3493 Kcal/kwh RPS 3314 Kcal/kwh 3167 Kcal/kwh IPGTPS 2450 Kcal/kwh* 2497 Kcal/kwh *Normative Approved by the Commission However, the Commission approved the station heat rate as under:

STATION 2005 - 06 2006 07 IPTPS 3235 Kcal/kwh 3235 Kcal/kwh RPS 3200 Kcal/kwh 3200 Kcal/kwh IPGTPS 2450 Kcal/kwh 2450 Kcal/kwh The appellant contends that the norms set out by the DERC were not achievable as the plants were very old and were poorly maintained by the erstwhile owners.

(c) Rebate on timely re-payment: DERC/Commission has allowed a rebate of 2% per month to the respondent No. 2 in case of timely payment of the bills for sale of electricity. The appellant says that the Commission has not taken into account the total outflow of cash caused by this 2% rebate. As per the Commission, the 2% rebate is matched by the interest allowed on the working capital for two months. As per the calculations of the appellant, there is a significant difference between the interest on working capital allowed for two months @ 10.25% per annum with monthly rest which works out to 20.5% per annum whereas the rebate of 2% per month works out to 24% per annum.

(d) Generation from RPH: The appellant says that the targeted PLF of RPH should not have been fixed at 60% in view of the long shut-down of the plant for rectification of the chronic problems of axial shift and vibration of unit Nos. 1 & 2 turbines.

(e) Variable cost of 1 PGTPS: The appellant claims that there was a mistake in calculating the fuel cost that should be allowed as pass through. The appellant wants fuel cost at 271.99 Crore calculated at station heat rate permitted by the Commission i.e. 2450 Kcal/kWh although actual heat rate achieved was 2426 Kcal/kWh. The Commission has allowed only 260.07 Crores on this account.

(f) PLF Target for recovery of fixed cost:

The Commission has fixed the station wise generation targets and calculated the fixed cost per unit based on the approved generation targets and in the concluding paragraphs the fixed cost recovery is tagged with target availability which was equivalent to target PLF fixed for payment & incentives. The appellant contends that target availability so fixed by the Commission for recovery of fixed cost is unreasonable and not according to facts and circumstances of the appellant and contrary to generation target approved by the order.

The Commission has fixed the following target availability for payment of incentives as below:

Station Target availability I.P. Station Rajghat Power Station GTPS 60 (combined for 2005-06 & 2006-07) If the target generation is taken into consideration the fixed cost has to be as per the following table:

Station Gross Generation (In MMU) PLF (%) Net Generation IPTPS 43.80 RPH 67.60 IPGTPS 60.70 The appellant contends a clarification of the issue is called for.

Response by DERC:

4. The DERC has filed a reply affidavit and has attempted to explain the rates in respect of each of the above items.

The responses are as under:

a) O&M Expenses: The Commission has allowed O&M expenses at Rs. 95.61 Crores only for the financial year 2005-06 based on norms recommended by CEA and the same is inclusive of extra allowances of Rs. 4 Crores towards insurance of plants. It is contended that the Commission has allowed increase of 4% over the O&M expenses for the financial year 2005-06. The Commission considers actual O&M expenses for the appellant to be on the higher side keeping in view of the parameters of operation which were considered by CEA while recommending normative O&M expenses for the financial year 2004-05 for these stations and after taking into consideration the vintage and sizes of the generation units.

b) Heat Rate: The Commission contends that so far as actual heat rate of IPTPS station is concerned, it has also mentioned in the impugned tariff order dated 22.09.06 that actual station heat rate of 3709 Kcal/kWh is very high on account of low operating levels and it needs to be improved. Due to non - availability of design heat rate data of the units of the IPTPS station, the Commission contends, it could not establish the reasonable heat rate in accordance with the provisions recommended by CEA and therefore, it retained the earlier approved figure of 3235 Kcal/kWh for IPTPS station which is inline with the draft PPA submitted by Transco. Similarly for the Rajghat Power House, the heat rate of 3314 Kcal/kWh is said to be high on account of low operating levels. This heat rate is said to be as per the PPA submitted by Transco. It is further contended by the Commission that the PLF / availability for the financial year 2005 - 06 as well as 2006 - 07 have been consciously relaxed but further relaxation was considered to be adverse to consumers interest. So far as IP Gas Turbine Station is concerned, the normative heat rate of 2450 Kcal/kWh for both the years 2005-06 and 2006-07 is said to be based on gross calorific value of gas. The appellant is entitled to the actual fuel cost incurred by it and not the fuel cost to which it would have been entitled to, had the station heat rate being same as was being prescribed by the Commission.

c) Rebate on timely payment: The Commission contends that rebate on timely payment is a trade off with interest on two months receivables considered in working capital requirement.

d) Generation from RPH and PLF targets for recovery of fixed cost: Coming to PLF factor, the Commission says it has considered the fact that major repairs were carried out by the appellant for the two units of RPH during 2005 - 06 and 2006 - 07 which had resulted in lower availability. However, given the positive result of enhanced generation at PLF above 80% after the repairs, the Commission clubbed the target availability for 2005-06 and 2006-07 together for recovery of full fixed cost of Rajghat Power Station.

e) Variable cost of IPGTP Station: The Commission explains that volume of Gas as submitted by the appellant was considered by it and that the difference in fuel cost was due to lower LNG cost which was provisionally considered by the Commission in arriving at variable cost of Rajghat Power house. Further it says that the LNG Gas cost has been worked out by using the heat of GCV for LNG Gas and conversion factor for converting the gas quantity to MMBTU. Consequently the difference in total fuel cost may be due to conversion factor adopted by the Commission, for converting the gas quantity to MMBTU. The Commission however, says that it has required the appellant to provide necessary details in support of its claims during the subsequent tariff filing and variation, if any, in actual fuel expense and approved fuel expense could be considered during the truing up of the expenses.

6. Additional Affidavit : The appellant has filed an additional affidavit to give certain facts. The following facts are mentioned in this additional affidavit: One of the generating stations namely IPTPS of the appellant is more than 38 years old and it was proposed to be shut-down due to vintage and other environmental factors but decision whether it should be shut-down was not final by the time the tariff petition was filed. However, Government of Delhi has decided to shut-down the IPTPS vide Cabinet decision No. 1204 dated 07.05.07. The appellant has also placed on record its own audited accounts and the CEA report on O&M expenses and proposed renovation and modernization works to be carried out by the appellant. The CEA after examining the various factors of the gas generating stations of the appellant gave the following conclusions and recommendations:

> 6.0 CONCLUSIONS & RECOMMENDATIONS:
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> i) The normative O&M expenses for the various plants of IPGCL, for the year 2004-05 are assessed as follows:
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> a) IP Station - 15 lakh/MW or Rs. 37.13 Crores
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> b) RPH Station- 17 lakh/MW or Rs. 22.95 Crores
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> c) GT Station - 10 lakh/MW or Rs. 28.20 Crores
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> ii) The actual expenses on various components of O&M expenses are higher for the year 2004-05 as compared to the above normative levels. In order to ensure continued plant operation, it is prudent that the normative O&M expenses may be considered as future bench marks to be achieved over a period of time, say 3-5 years. Till then, it is recommended that well audited actual expenses may form the basis for 2004-05 with yearly targeted improvement for future.
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> iii) There shall be specific commitment for targeted improvement in future.
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> iv) The large employee strength is a matter of concern as it is resulting in large outgo of funds towards salary etc. There is need for restructuring/redeployment of existing staff in a more productive manner for which a separate study can be conducted by an expert group/consultant.
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> v) The accounting procedure needs to be more organized with well defined heads & sub-heads for various components of O&M expenses.
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> vi) The assessment is recommendatory in nature and within the framework of constraints/limitations of the data furnished.

7. The Directors report for the financial year ending 31st March, 2007 submitted by the appellant mentions the operation performance of the different stations as under :

Operational Performance S. No. PARTICULARS IP STN.

RPH GTPS TOTAL Capacity (MW) 247.5 664.5 Gross Generation(MU) 952.31 634.92 1412.22 2999.45 PLF (%) 43.92 53.69 57.17 51.53 Auxililary Consumption (%) 15.51 12.10.

2.80 Energy sent out (MU) 804.59 558.07 1372.33 2734.99 During the year, the PLF of I.P.Station has marginally decreased from 45.42% to 43.92% due to the policy decision of not incurring of major expenditure on Repair & maintenance works, in view of its proposed closure in the year, 2010. The generation from Unit No. 4 also remained low due to overhauling of Unit followed by condenser problem PLF of Rajghat Power Station increased from 48.57% to 53.69%. However, due to shut down of Unit No. 1 for more than three months to attend to the problems of high vibration axial shift in turbine, persisting for last several years, failure of turbine blade in Unit 2 in December 2006 and damaging of bearings of turbine of Unit I on account of lube oil problem, the target set by DERC could not be achieved.

The PLF of GT station had decreased from 70.76% to 57.71% mainly on account of (a) backing down to the tune of 138.6 MU of generation by SLDC (b) non availability of gas due to flooding of GAIL terminal, Hazira, shut down at ONGC station in January, 07 and low allocation of gas resulted in loss of 113.17 MU of generation (c) break down of generator of GT 6 while re-commissioning after overhauling by M/s Alstom, the OEM (d) prolonged shut down of STG-1 for its overhauling by M/s. BHEL, the OEM and (e) break down of STG II wherein internals of STG and its rotor stages were damaged due to dislodging of balancing in the machine.

8. The appeal is also opposed by two distributing companies namely BSES Yamuna Power Ltd. and BSES Rajdhani Power Ltd. BSES Yamuna Power Ltd., respondent No. 4, disputes the vintage of RPH and IPGTPS. According to this respondent the accumulated depreciation being only 50% of the GFA, there is clear indication that the appellant has carried out refurbishment and modernization work after the date of commissioning of the station and therefore the plant cannot be treated as old. Further this respondent says that station heat rate, auxiliary consumption, specific consumption of fuel, PLF should be allowed in line with CEA norms. Coming to station heat rate it is alleged by BSES Yamuna that thermal power stations like Tanda have been allowed heat rate of 3000 Kcal/kWh and therefore, the same norms should be prescribed for the appellant. This respondent is opposed to increasing the heat rate and contends that any relaxation in heat rate would result in inefficiency and consequent burden on the distributing company and the consumers.

9. North Delhi Power Ltd., respondent No. 5, has also opposed the appeal. It contends, inter alia, that the heat rate for RPH approved by the Commission is higher than what was sought and the heat rate approved for IPGPS is at 2450 is quite reasonable. It supports the decision of 2% rebate on timely payment on the ground that this will be compensated by the interest that would be allowed on working capital and that such rebate is in accordance with CERC norms 2004 and is being allowed to all central sector generating companies. The NDPL supports the views of the Commission and all other aspects as well.

10. Decision with reasons:

> O&M Expenses: As mentioned in Paragraph 4(a) above, the Commission has strictly adhered to the normative increase as suggested by the CEA. The petitioner requested the Commission to ask the CEA to re-consider the report. The Commission, however, went by the CEA report which according to the Commission had been prepared after taking all relevant parameters of operation into consideration. The Commission, however, was uncertain as to how long the IP station can be operated based upon various conditions imposed by statutory authority / courts etc. The Commission therefore did not relax the norms. However, the Commission did not consider if it was proper to give any further time to the appellant to improve its performance for the purpose of reducing cost.

11. Reverting to the report of the CEA, we find that the authority itself never meant that the norms prescribed by it be strictly taken for compliance. In recommendation No. VI, the authority itself says that the assessment is recommendatory in nature and within the framework of constraints / limitations of the data furnished. The Commission feels satisfied with applying the CEA norms and also feels compelled to do so for otherwise the interest of consumers will suffer. The Commission is duty bound to allow all reasonable O&M expenses as pass through. The Commission is allowed to only impose a prudence check. If the O&M expense had escalated within the norms prescribed by CEA, the task of the Commission would have been quite easy. However, in the present case, the O&M expenses have gone higher than the CEA norms. The Commission, therefore, was required to examine the expenditure incurred by the appellant for various purposes and to detect if the appellant had incurred any avoidable expense. The appellants are only successors in interest of the erstwhile DVB and it has inherited some old power plants. It, along with power plant, has also received a large number of employees which in the present legal regime cannot be shed immediately. The appellant has attempted to reduce the number of employees by offering VRS. The plant being old the maintenance expense can also be higher than what is estimated at the given point of time. The fact that the appellant has been making efforts to improve its performance is clear from the figures of actual heat rate for 2005-06 and 2006-07 which show that there is a fall in the station heat rate in 2006 compared to in the year 2005. It cannot be said that the appellant had altogether been irresponsible in its expenditure. Therefore, it will be appropriate for the Commission to examine individual items of expenditure and disallow only those which it finds as avoidable or imprudently high. We had an occasion to examine the concept of prudence check in the case of NDPL v. DERC Energy Law Reports (APTEL) 2007 193. The Commission had allowed an A&G expenses for distribution companies of Delhi only to the escalation of 4% over the previous year's expenses. The Commission had made deduction on account of legal expenses and other general expenses. We held that the Commission would have to allow such expenses which are justifiable and can disallow such expenditures which were not justified. The Commission cannot simply apply the normative rates of escalation and feel that its function of regulation is thereby over. While the Commission is duty bound to regulate the generation, transmission and distribution keeping in view the interest of consumers, it is also bound to see that the generator, transmitter and distributor gets a fair return, over and above the expenses. We, therefore, have no option but to hold that the Commission has to do some more exercise in arriving at the correct figure of O&M expenses

which can be taken as pass through in tariff. It has to examine individual items of expenditure and reject those which were clearly avoidable or imprudent or impermissible and allow the rest as pass through.

12. Heat Rate: The petitioner could not submit the design heat rate for IP station which was nearly 38 years old. The CEA norms, for station heat rate is based on the design heat rate. Therefore, the Commission could not employ the CEA norms for the station heat rate of the IPTPA station. Accordingly the Commission retained the approved figure of 3235 Kcal/kWh which was agreed to by the petitioner inline with the draft PPA submitted by Transco along with ARR petition for 2004-05. The petitioner submitted before the Commission that in order to comply with the directions of Delhi Pollution Control Committee IPTPA station was proposed to be closed down and therefore no R&M expenses could be taken for improvement or even for maintaining the same station heat rate. We are informed during arguments that the final decision to close down was taken after the end of 2006-07. The final closing will be in 2010. In view of this situation, it will only be fair for the Commission to bear with the station heat rate which the appellant has been able to achieve for this station during the period in question.

13. For the RPH station the appellant has actually been able to achieve the given target of station heat rate. While target was 3200 Kcal/kWh the appellant has been able to achieve 3167 Kcal/kWh.

14. For IPGTPS, the target of 2450 Kcal/kWh could not be achieved. Actual heat rate for 2006-07 was 2497 Kcal/kWh. So far as IPGTPS is concerned, the Commission has fixed the station heat rate norm as 2450 Kcal/kWh for financial years 2005-06 and 2006-07. This is based on the gross calorific value of gas. The appellant pleads that the station was very poorly maintained and that the availability of gas was greatly reduced during the period in question.

15. The impugned order shows that the petitioner had sufficiently canvassed its case of shortage of gas caused by the cuts imposed by GAIL. The Commission has not analysed in the impugned order the affect of such cuts on the station heat rate of the IPGTPS station. Even if the other factors mentioned in the 'Director's report' above are ignored the shortage of gas should have been taken into account by Commission because this is not within the control of the appellant. We, therefore, feel that the Commission needs to carry out this exercise afresh so far as the station heat rate of IPGTPS is concerned. The Commission will now refix the target heat rate for the IPGTPS from 2006-07 after taking into consideration the shortage of gas as well as the factor mentioned in the Directors report as indicated in para 7 above. Consequent benefit be given to the appellant in the truing up and in the subsequent tariff order.

16. Rebate on timely repayment: So far as rebate on timely repayment is concerned, the Commission feels that the cash outflow on this account is sufficiently matched by the interest allowed on working capital. The appellant submits that the interest allowed by the Commission on working capital works out to 20.50 % p.e. while the rebate worked out to 24% p.a. As such the trade off concept has caused a loss of Rs. 1.94 Crores annually. The appellant accordingly says that this amount of Rs. 1.94 Crores should be allowed to be recovered from tariff. This point was also submitted in the review petition filed by the appellant. The Commission in the review order dated 30th March, 2007 has noted the

submission but has not taken any decision on the submission and has merely reiterated the original order dated 22nd September, 2006. We have given our thought on the subject and we feel that there is no reason why the appellant should lose the sum of Rs. 1.94 Crores annually because of the rebate allowed to the transmission licensee for timely payment. The Commission has to allow the sum of Rs. 1.94 Crore as pass through and the same be done in the truing up and subsequent tariff orders.

17. Generation from Rajghat Power House, PLF target and recovery of fixed cost: For recovery of fixed cost the plant load factor for Rajghat power house has been held to be 60%. The appellant contends that it has not been able to achieve this target due to long shut-down of one of its plants due to axial shift and variation. The necessary rectification has been carried out. The Commission has considered this aspect. The Commission found that after the rectification combined PLF of both the units has risen very high so much so that during the first two weeks of November, 2006 the PLF of the two units was 96.56%. The Commission says that given the positive results of enhanced and reliable generation, PLF of 80% after repairs, the Commission clubbed the target availability of financial year 2005-06 and 2006-07 together for recovery of full fixed cost of Rajghat Power house. Even in the review order the Commission reiterated the desired target of 60%.

18. As per the operation norms laid down in Regulation 16 of Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2004, while the recovery of full capacity (fixed charges) is a function of target availability whereas incentive is payable if PLF exceeds the target PLF. The aforesaid Regulation reads as under:

> 16. Norms of operation: The norms of operation as given hereunder shall apply:
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> (i) Target Availability for recovery of full Capacity (Fixed Charges)
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> (a) All thermal power generating stations, except those covered under Clauses (b) and (c) below -80%
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> (b) Thermal power generating stations of Neyveli Lignite Corporation Ltd. (TPS-I, TPS-II, Stage I & II and TPS-I Expansion) and Talchar Thermal Power Station of National Thermal Power Corporation Ltd. -75%.
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> (c) Tanda Thermal Power Station of National Thermal Power Corporation Ltd. -60%
> Note:
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> Recovery of capacity (Fixed) charges below the level of target availability shall be on pro rata basis. At zero availability, no capacity charges shall be payable
>
> ii) Target Plant Load Factor for Incentive
>
> (a) All thermal power generating stations, except those covered under Clauses (b) and (c) below - 80%

(b) Thermal power generating stations of Neyveli Lignite Corporation Ltd. (TPS-I, TPS-II, Stage I & II and TPS-I Expansion) and Talcher Thermal Power Station of National Thermal Power Corporation - 75%

(c) Tanda Thermal Power Station of National Thermal Power Corporation Ltd. - 60%

19. Delhi Electricity Regulatory Commission's (Terms and Conditions for Determination of Generation Tariff) Regulations 2007 also at Regulation 7.3 provides the target availability for recovery of full fixed charges and target PLF for claiming incentives. It may be pointed out that if the station is backed down under the instructions of SLDC it will be considered as deemed available in so far as the recovery of the fixed charges is concerned. In case if availability is lower than the target availability the fixed cost admissible is reduced proportionately with no fixed charges allowed at zero availability. Also incentive for energy generation is permitted when station achieves PLF higher than the target PLF. The target availability for the recovery of fixed charges in the instant case has been fixed equivalent to target PLF for payment of incentives. Therefore, if the station has exceeded the target PLF it will not only be eligible for recovery of full fixed charges but also for incentives. The appellant can neither claim recovery of full fixed charges nor incentive based on the PLF performance as long as it does not achieve the PLF target and exceeds it for incentive payment. In the instant case the gross generation as projected by the appellant to the Commission cannot be used to determine the target PLF but it will only represent the quantum of generation from each station for the purpose of energy balancing. The PLF so derived cannot be used as target PLF. The target availability of a station is based on the declared capacity during the year and not on the PLF achieved as claimed by the appellant.

20. Taking the case of RPH station the Commission has considered the vintage and current status of plant operation including long shut downs of unit Nos. 1 and 2 and has relaxed the target availability for the recovery of fixed charges of RPH to 60% combined for FY 2005-06 and 2006-07 as against 70%. It is to be noted that practically one unit of the station was not available for operation because of major repairs for more than 8 months out of 24 months in FY 2005-06 and 2006-07. Viewing it year-wise, the station was having one unit with chronic problem and the second unit was available for only 4 months after the repairs in FY 2005-06. Whereas in FY 2006-07 one unit was available after repairs for the entire year and the second unit was only available for nearly 7 months after repairs. Expected availability of RPH in FY 2005-06 is widely different from that achievable in FY 2006-07 and combining them and fixing relaxed normative target availability of 60% will not represent true picture. It will be reasonable to fix target availability separately for each year. We, therefore, direct the Commission to fix target availability of RPH for recovery of fixed cost separately for each year while taking into account the factors for relaxation. No incentive, however, is allowed if the PLF does not exceed target PLF of 70%.

21. So far as variable cost of IPGPT station is concerned, the Commission has taken note of the possibility of an error. In the reply affidavit the Commission says that it asked the appellant to provide necessary details in support of its claim during the subsequent tariff filings and assured the appellant that the Commission would consider variation, if any, between actual fuel expense and fuel expense approved during the truing up exercise.

22. The appeal is accordingly allowed in part with the directions indicated in Paragraphs 11,12,15,16 & 20 above. Since much time has passed from the end of the financial year 2006-07, we direct that instead of redetermining the tariff for that year, the effect of reconsideration / recalculation shall be given in the truing up exercise and consequent tariff in the next tariff period.

Pronounced in open court on this 10th day of January, 2008.